

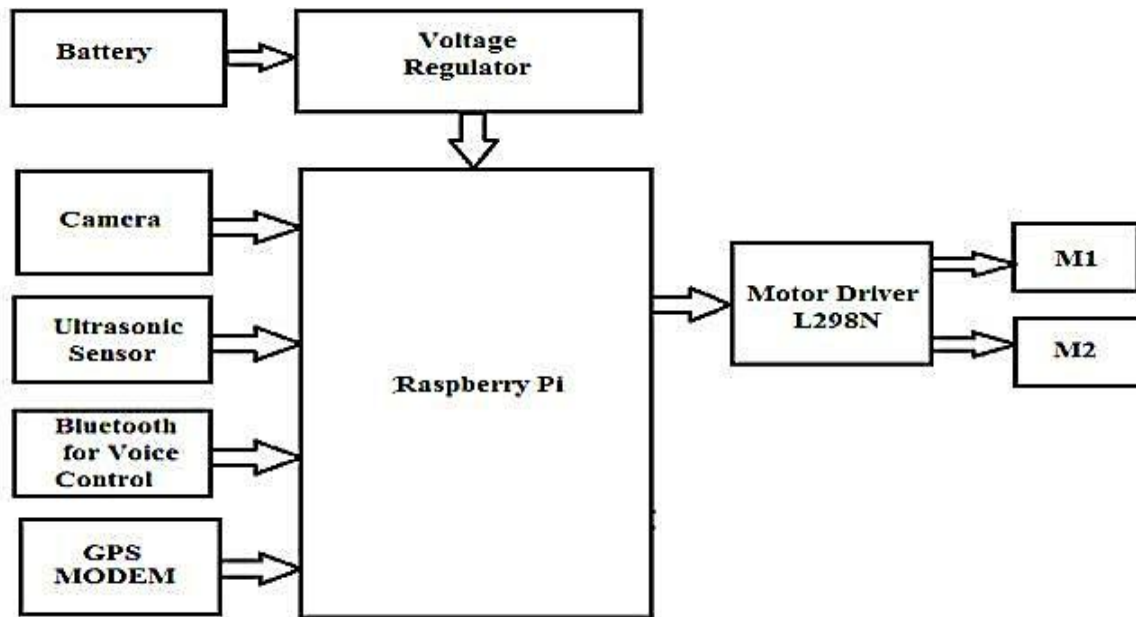


Figure 1: Block diagram of autonomous car.

Using CNN, we take inputs from the RPI camera which sends real-time video to the RPI board. These videos are divided into frames and fed into the algorithm. Once the CNN has been trained with the training Dataset, it can interpret the traffic signals. The CNN consists of three main layers: input layer, hidden layer and output layer. The first layer is the convolutional layer where the input is given. Here the image input is always in the matrix form that is, a filter is applied to an image and the pixels are recorded into matrix form. In the hidden layers, the image undergoes through many filters and becomes a linear pixel array. The output is according to the linear pixel array.

We then detect the road or lane from Open CV library. Here we take the real time video frames from the RPI camera as the input for the algorithm. First we convert the image into BW image by adding a gray scale which gives us a filtered image frame. Next this filtered image to plot the Hough lines on the lane marking. Using Hough lines conversion we can plot the perfect lane markings or we can detect the edges of the road. To find the curve or turns on the road, we divide the Hough

line plotted image into three major ratios: 2:2:5. Next we draw imaginary Hough line on to the centre of the image which depicts the straight line. Next we calculate the angle of the plotted Hough line with respect to the imaginary plotted Hough line. This angle gives us the perfect steering ratio for the RC car to make a turn. We use the white-line markings on the side of the roads as our base lane for the lane detection algorithm.

Ultrasonic sensor module is used to calculate the distance between the obstacle and our RC car. The ultrasonic sensor which is placed in front of the RC car emits a low frequency ultrasonic waves which hit the obstacle in front of us and rebounds back to the sensor. The sensor captures the rebounded ultrasonic waves and calculates the distance between the obstacles using the formula:

$$\text{Distance} = (\text{speed} \times \text{time}) / 2$$

Using output of lane detection algorithm which gives us the steering angle, Haar cascade algorithm gives us the stop